Global Rectification and Decoupled Registration for Few-Shot Segmentation in Remote Sensing Imagery

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***Abstract*— Few-shot segmentation (FSS), which aims to determine specific objects in the query image given only a handful of densely labeled samples, has received extensive academic attention in recent years. However, most existing FSS methods are designed for natural images, and few works have been done to investigate more realistic and challenging applications, e.g., remote sensing image understanding. In such a setup, the complex nature of the raw images would undoubtedly further increase the difficulty of the segmentation task. To couple with potential inference failures, we propose a novel and powerful remote sensing FSS framework with global rectification (GR) and decoupled registration (DR), termed R^{2}Net. Specifically, a series of dynamically updated global prototypes are utilized to provide auxiliary nontarget segmentation cues and to prevent inaccurate prototype activation resulting from the variability between query-support image pairs. The foreground (FG) and background information flows are then decoupled for more targeted and tailored object localization, avoiding unnecessary confusion from information redundancy. Furthermore, we impose additional constraints to promote interclass separability and intraclass compactness. Extensive experiments on the standard benchmark iSAID-5^i demonstrate the superiority of the proposed R^{2}Net over state-of-the-art FSS models. The code is available at https://github.com/chunbolang/R2Net.**

***Index Terms*— Few-shot learning, few-shot segmentation (FSS), meta-learning, remote sensing, semantic segmentation.**

NOMENCLATURE

Notation	Description
$\mathbf{X}$; $\mathbf{Y}$	Image; mask.
$\mathbf{F}$	Feature representation.
$\mathbf{M}$	Base maps.
$\mathbf{p}$	Prototype.
s; q	Support; query.
k	Index of shots.
$\mathcal{P}$	Prototype set.
$\mathcal{G}$	Global average pooling.
$\mathcal{C}$	Cosine similarity.
$\mathcal{A}$	Argmax operation.

Manuscript received 3 March 2023; revised 16 June 2023; accepted 30 July 2023. Date of publication 2 August 2023; date of current version 16 August 2023. This work was supported in part by the National Science Foundation of China under Grant 62136007 and Grant 62036005, in part by the Natural Science Basic Research Program of Shaanxi under Grant 2021JC-16 and Grant 2023-JC-ZD-36, in part by the Innovation Foundation for Doctor Dissertation of Northwestern Polytechnical University under Grant CX2023014, and in part by the Cultivation Foundation for Excellent Doctoral Dissertation of the School of Automation of Northwestern Polytechnical University. *(Corresponding author: Gong Cheng.)*

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Digital Object Identifier 10.1109/TGRS.2023.3301003

I. INTRODUCTION

REMOTE sensing images contain a wealth of geographical information, and their intelligent interpretation provides crucial support for scientific research, resource management, and decision-making [1], [2], [3], [4], [5]. Semantic segmentation, as an essential component, strives to allocate each pixel within an image to one of the predefined geospatial semantic categories [6], [7]. With the development of modern remote sensing sensors, such a vision task has received extensive attention and enjoys broad application prospects, e.g., urban planning [8], environmental monitoring [9], and natural disaster detection [10]. In recent years, the emergence of deep learning techniques has dramatically contributed to the accuracy and reliability of segmentation results, pushing the field to new heights [11]. However, even this promising technique is not perfect. Collecting a substantial number of annotated samples seems to gradually become a necessity for training deep models, which not only places tremendous burdens but also compromises the ability to recognize new classes [12], [13], [14], [15].

Against that backdrop, few-shot segmentation (FSS) follows into the spotlight, dedicated to parsing task-relevant targets in query images by relying on a handful of labeled support data [16]. Since the task setting of FSS was first established by Shaban et al. [17], many researchers have successively provided valuable insights and condensed them into methods. Zhang et al. [18] designed a segmentation model with an iterative optimization module and an attention mechanism, in which the prediction results are iteratively refined and the information from multiple support samples is efficiently fused. Another essential feature of the proposed model, which inspired most of the follow-up work, was the utilization of the frozen backbone network to improve generalization capabilities. Tian et al. [19] analyzed in depth the active role of prior knowledge and spatial consistency on the current task, and accordingly proposed the training-free mask acquisition method and feature enrichment module. Despite considerable progress, most current work in the field has focused on natural image processing, with few attempts at other, more realistic applications [20], [21], [22]. Taking the remote sensing images targeted in this article as an example, they have greater interclass similarity and intraclass diversity than natural images. As shown in the first row of Fig. 1, closely arranged ships, vehicles, planes, etc., have high interclass similarity under top-view imaging conditions, especially the former two classes are easy to be confused in appearance. The second row indicates the diversity between different objects in the same

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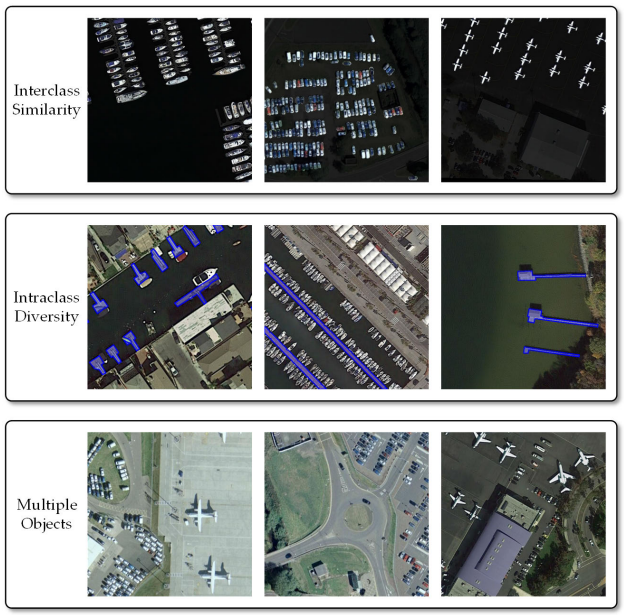


Fig. 1. Different characteristics of remote sensing imagery, including interclass similarity, intraclass diversity, and multiple objects, are shown from top to bottom, respectively, posing further challenges for semantic segmentation in the low-data regime.

class (harbor), i.e., with few ships around, lots of ships around, and no ships around. In addition, there are often multiple densely arranged instances in the original image that need to be segmented, as illustrated in the last row of Fig. 1. These properties undoubtedly increase the difficulty of the task, enabling previous frameworks to produce unsatisfactory segmentation results, such as irrelevant object activation and imprecise object boundaries.

In order to cope with the obstacles mentioned above, we specifically design an FSS network with global rectification (GR) and decoupled registration (DR) for remote sensing images in this work, termed R^2Net. Specifically, the local prototypes computed by masked average pooling are rectified by a set of global prototypes that can be dynamically updated. In this way, nontarget auxiliary segmentation cues are complemented to facilitate inference, and inaccurate prototype activation due to discrepancies between query-support image pairs is avoided. Considering that the direct stacking of heterogeneous information may create confusion, the foreground (FG)- and background-related cues embedded in the rectified prototype are then decoupled for more targeted and tailored object localization. Moreover, for the complex characteristics of remote sensing images, we introduce additional constraints to promote interclass separability and intraclass compactness, driving the few-shot model to extract robust and discriminative feature representations during optimization.

Our main contributions can be summarized as follows.

1) We specifically devise a novel framework for FSS in remote sensing imagery, termed R^2Net, which dramatically reduces the burden of sample annotation and facilitates the performance of the segmentation model in low-data regimes.
2) Our proposed R^2Net has two main components: i.e., GR and DR. The former leverages a set of global prototypes that can be dynamically updated to rectify the local ones, while the latter decouples the FG and background information flows for more tailored object localization.
3) We conduct extensive experiments on the standard benchmark to validate the effectiveness of our framework and achieve state-of-the-art performance.

The remainder of the article is organized as follows. Section II reviews the representative work in related fields. Section III describes the methodology in detail and presents an overall flowchart. The analysis and discussion of the experimental results are given in Section IV. Finally, Section V summarizes the contributions and experimental findings of the article.

II. Related Work

A. Generic Semantic Segmentation

Semantic segmentation aims to assign one of the predefined semantic categories to each pixel in an image, and it has been widely investigated by researchers as an important technique for fine-grained scene understanding [23]. Long et al. [6] were the first to design fully convolutional neural networks (FCNs) for the semantic segmentation task, which have been widely adopted by researchers due to a variety of user-friendly features, such as strong representational capabilities and end-to-end training paradigms. Fu et al. [24] proposed a dual attention network (DANet) to model semantic interdependencies in the spatial dimension and channel dimension, respectively, where the local features are adaptively integrated with their global dependencies. Huang et al. [25] applied a novel yet effective criss-cross attention module at each spatial location that collects rich contextual cues for all pixels on the criss-cross path. In addition, semantic segmentation methods based on deep learning technologies have also dominated and made unprecedented progress in the field of remote sensing image interpretation [26], [27]. However, the aforementioned methods rely heavily on large-scale datasets for training, which is resource-intensive and difficult to handle unseen categories.

B. Few-Shot Learning

Few-shot learning, with its unique advantage of dealing with unseen categories, has gradually become a research hotspot in the machine learning community [28], [29], [30]. The main research goal of this technique is to recognize unseen concepts given a small annotated sample, i.e., prior knowledge. Vinyals et al. [31] proposed an episode-based training paradigm in which the original dataset is divided into multiple sets of episodes, i.e., learning tasks, for model training. Such an episodic learning approach is closely related to meta-learning and seeks to improve the generalization ability of the few-shot model by simulating evaluation scenarios during the training procedure. Snell et al. [32] introduced the idea of prototype learning into the field of few-shot learning, where each category is represented by a specific feature vector (i.e., prototype) and distinguished in the well-constructed high-dimensional feature space. In addition, many few-shot learning

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methods specific to the field of remote sensing image processing have also been proposed in recent years [33], [34]. We note that most of the research has focused on scene classification and less on higher-level scene understanding tasks such as semantic segmentation.

C. Few-Shot Segmentation

FSS can be simply understood as an extension of few-shot learning techniques in the field of semantic segmentation, and its fundamental goal is to segment potential objects within the image to be tested given several densely labeled images [35], [36], [37]. The setting of this particular semantic segmentation task was first established by Shaban et al. [17], and a network with conditioning and segmentation branches was proposed to perform dense pixel-level predictions. Building on this landmark work, many researchers successively provided valuable insights, which were further condensed into methods. The CANet designed by Zhang et al. [18] came with an iterative optimization module and an attention mechanism, in which the prediction results are iteratively refined and the information from multiple support samples is efficiently fused. Another important characteristic of CANet was the use of frozen backbones to boost generalization, which inspired most of the subsequent work. In order to incorporate the category information embedded in the support images into the few-shot network, Zhang et al. [38] proposed the masked average pooling operation, which aggregates the features corresponding to the task-relevant FG regions. Such a prototype generation strategy has been frequently adopted in subsequent approaches due to its broad applicability and high efficiency. Tian et al. [19] analyzed in depth the active role of prior knowledge and spatial consistency on the current task, and accordingly proposed the prior guided feature enrichment network (PFENet) with training-free mask acquisition method and feature enrichment module.

However, the vast majority of FSS studies currently focus on natural image processing, and few of them are specifically designed considering the complex characteristics of remote sensing images. Yao et al. [21] proposed a scale-aware FSS network for aerial images, where the multiprototype spirit and focal loss function facilitate the segmentation performance of the network. Wang et al. [22] designed a metametric learning framework, called DMML-Net, for few-shot geographic object segmentation, where the deep feature pyramid comparison sub-network, affinity-based fusion mechanism, and metametric learning paradigm effectively improve the segmentation accuracy. For this reason, we devise a novel yet powerful remote sensing FSS framework and validate its unique advantages over previous approaches.

III. Methodology

A. Task Description

The goal of FSS in remote sensing imagery is to identify potential geospatial objects within query images relying on only a small number of densely labeled samples, which still follows the setting of standard FSS [17]. Specifically, we construct a training dataset $\mathcal{D}_{\text{train}}$ and a test dataset $\mathcal{D}_{\text{test}}$ based on two *disjoint* category sets C_{train} and C_{test}. Each image in $\mathcal{D}_{\text{train}}$ contains at least one category of C_{train}, and the same is true for $\mathcal{D}_{\text{test}}$.

Meta-learning paradigms are often employed to promote generalization, in which a series of *episodes* (i.e., subtasks) are randomly sampled from $\mathcal{D}_{\text{train}}$ to simulate the few-shot scenario during testing [31]. Taking the K-shot task as an example, we usually select $K+1$ image-mask pairs to compose the metadata for mini-batch optimization, denoted as $\mathcal{E}=\{(\mathbf{X}_s^k,\mathbf{Y}_s^k)\}_{k=1}^K\cup\{(\mathbf{X}_q,\mathbf{Y}_q)\}$ where the two included subsets represent the support set $\mathcal{S}$ and the query set $\mathcal{Q}$, respectively. In each forward propagation, the support set $\mathcal{S}$ and the query image $\mathbf{X}_q$ are fed into the segmentation model to produce the prediction result $\hat{\mathbf{Y}}_q$. The query mask $\mathbf{Y}_q$ is used to compute the loss during the meta-training phase but is not available during the meta-testing phase. Note that the majority of current FSS models focus on the 1-way K-shot task, and thus both the support mask and the query mask indicate the objects of a particular geospatial category in each episode. For ease of reading, Nomenclature summarizes the notations involved in this article and their corresponding descriptions.

B. Method Overview

The framework of our proposed R^{2}Net is shown in Fig. 2, which has two important components: GR and DR. Given a one-shot learning episode $\mathcal{E}$ with support set $\mathcal{S}=\{(\mathbf{X}_s,\mathbf{Y}_s)\}$ and a query set $\mathcal{Q}=\{(\mathbf{X}_q,\mathbf{Y}_q)\}$, the support and query images are first fed separately into a weight-sharing backbone network, and the original features $\{\mathbf{F}_s^{\text{ori}},\mathbf{F}_q^{\text{ori}}\}$ are produced accordingly. Then we implement a base layer $\mathcal{B}(\cdot)$ and a meta layer $\mathcal{M}(\cdot)$ to extract base maps $\mathbf{M}_*$ and meta features $\mathbf{F}_*$, respectively, where $*\in\{s,q\}$ indexes the support and query branches. With heterogeneous region descriptors $\{\mathbf{M_s},\mathbf{Y}_s\}$, we further apply a global average pooling (GAP) operation $\mathcal{G}(\cdot,\cdot)$ on support meta features $\mathbf{F_s}$ to obtain local prototypes $\mathbf{p}_{\text{local}}$. Considering that the category representations generated by a single episode are not sufficiently robust, the local prototypes $\mathbf{p}_{\text{local}}$ are *rectified* by several dynamically updated global prototypes $\mathbf{p}_{\text{global}}$. The segmentation cues embedded in the merged prototypes $\mathbf{p}_{\text{merge}}$ are subsequently decoupled according to their properties to specifically *register* the FG- and background-related regions of query features $\mathbf{F_q}$. Finally, the refined query features $\mathbf{F}_q^{\text{out}}$ are fed into the segmentation head with the atrous spatial pyramid pooling (ASPP) module and recoupled to yield prediction results.

C. Meta and Base Layers

Given the original features $\{\mathbf{F}_s^{\text{ori}},\mathbf{F}_q^{\text{ori}}\}$ output from the backbone network, we first introduce a base layer $\mathcal{B}(\cdot)$ and a meta layer $\mathcal{M}(\cdot)$ to generate the base maps $\{\mathbf{M_s},\mathbf{M_q}\}$ and metafeatures $\{\mathbf{F_s},\mathbf{F_q}\}$, respectively. The former indicates the probability maps regarding base categories C_{train}, while the latter serves as an intermediate feature to yield the prediction about the current category that represents the base category $c\in C_{\text{train}}$ during *meta-training* phase and the novel category $c\in C_{\text{test}}$ during *meta-testing* phase.

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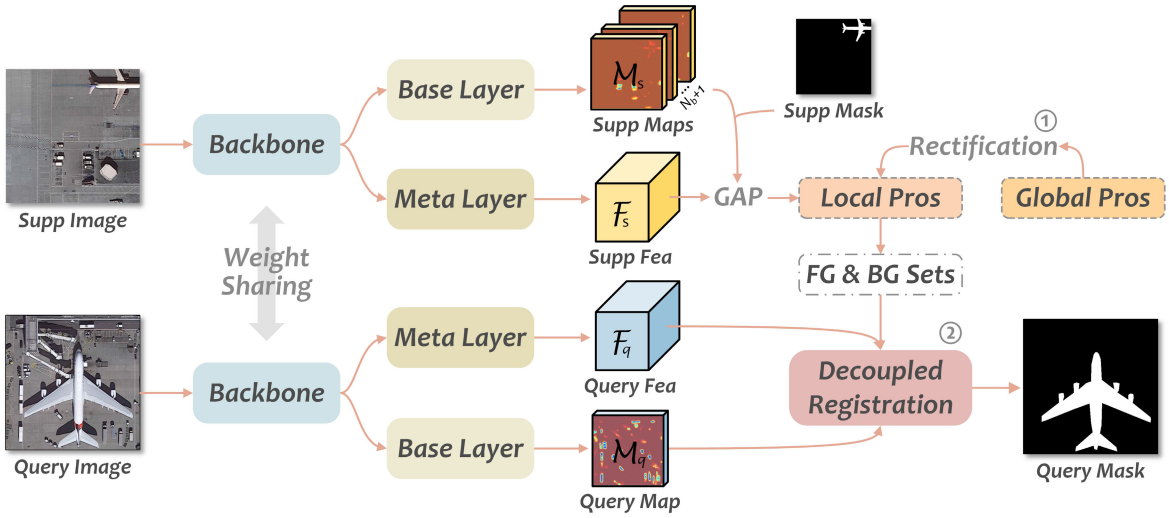


Fig. 2. Overall architecture of the proposed method. The support and query images are first fed separately into a weight-sharing backbone network. Then, the base and meta layers are employed to generate the base maps $\{\mathbf{M_s}, \mathbf{M_q}\}$ and metafeatures $\{\mathbf{F_s}, \mathbf{F_q}\}$, respectively. With efficient GAP, the local prototypes $\mathbf{p}_{\text{local}}$ are computed and further rectified by global prototypes $\mathbf{p}_{\text{global}}$. Finally, the segmentation cues embedded in the merged prototypes $\mathbf{p}_{\text{merge}}$ are decoupled to specifically mine FG- and background-related objects. Circled digits ① and ② correspond to the proposed GR (Fig. 3) and DR (Fig. 4) modules, respectively.

In this way, the potential objects in the support and query images can be well identified, especially the base-class objects that tend to affect the segmentation accuracy of the current task. Taking the *support branch* as an example, the above feature extraction process can be defined as

$$\mathbf{M_s} = \mathcal{B}\left(\mathbf{F}_s^{\text{ori}}\right) \in \mathbb{R}^{(N_b+1)\times H\times W} \tag{1}$$

$$\mathbf{F_s} = \mathcal{M}\left(\mathbf{F}_s^{\text{ori}}\right) \in \mathbb{R}^{C\times H\times W} \tag{2}$$

where $\mathcal{B}(\cdot)$ and $\mathcal{M}(\cdot)$ indicate the base layer and meta layer, respectively. We follow the decoder of [39] to construct $\mathcal{B}(\cdot)$, while $\mathcal{M}(\cdot)$ is a plain 1×1 convolution operation for dimensionality reduction. Note that the base maps $\mathbf{M_q}$ and meta features $\mathbf{F_q}$ of the query branch are acquired similarly. $N_b = |C_{\text{train}}|$ denotes the number of base categories, and $N_b = 10$ for the iSAID-5^i dataset used in this work.

D. Global Rectification

After determining the heterogeneous objects within the support image, we leverage the GAP operation $\mathcal{G}(\cdot,\cdot)$ to obtain condensed representations, i.e., the local prototypes in the upper left corner of Fig. 3. The above procedure can be mathematically defined as follows:

$$\mathbf{p}_{\text{local}} = \mathcal{G}\left(\mathbf{M_s} \oplus \tilde{\mathbf{Y}}_s \oplus \left(1-\tilde{\mathbf{Y}}_s\right), \mathbf{F_s}\right) \in \mathbb{R}^{(N_b+3)\times C} \tag{3}$$

where $\oplus$ represents the concatenation operation along channel dimensions, and $\tilde{\mathbf{Y}}_s \in \{0,1\}^{H\times W}$ indicates the *down-sampled* support mask using nearest neighbor interpolation to match the spatial resolution of the feature map.

Considering the limitations and incompleteness of the representations derived from individual tasks, we utilize a set of dynamically updated global prototypes $\mathbf{p}_{\text{global}}$ to rectify the local ones $\mathbf{p}_{\text{local}}$. And these global prototypes, which are randomly initialized, can be optimized with model training, gradually becoming general-purpose descriptors covering a wide range of tasks. The procedure of prototype rectification can be defined as follows:

$$\mathbf{p}_{\text{merge}}^{(i)} = \alpha \cdot \mathbf{p}_{\text{local}}^{(i)} + \beta \cdot \mathbf{p}_{\text{global}}^{(i)} \tag{4}$$

where $\mathbf{p}_{\text{merge}}$ denotes the merged prototypes obtained by rectification, and $i \in \mathcal{I} = \{0, \ldots, N_b\}$ indexes the elements in each prototype set. Note that the first $N_b + 1$ elements in $\{\mathbf{p}_{\text{local}}^{(j)}\}_{j\in\mathcal{I}'}$ correspond to all the elements in $\{\mathbf{p}_{\text{global}}^{(i)}\}_{i\in\mathcal{I}}$, where $\mathcal{I}' = \{0, \ldots, N_b+2\}$; while the last two elements in $\{\mathbf{p}_{\text{local}}^{(j)}\}_{j\in\mathcal{I}'}$ correspond to the FG and background prototypes of the current task. α and β represent the merging coefficients, which can be set as hyperparameters or learnable parameters.

These prototypes are then decoupled according to their FG and background attributes, where the prototypes with the same attributes form a new set. Thus, the FG set $\mathcal{P}_{\text{FG}}$ and the

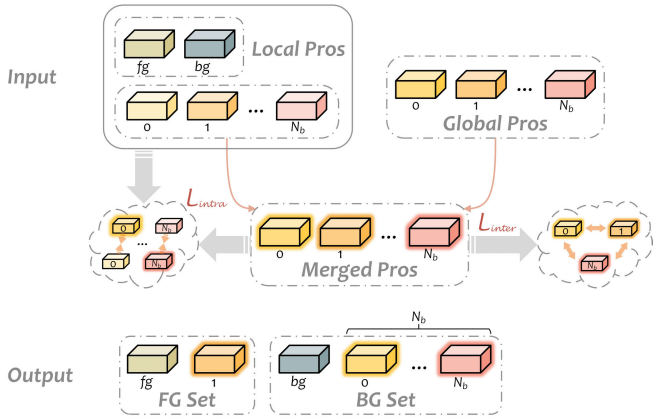


Fig. 3. Pipeline for GR during the *meta-training* phase. Given the computed local prototypes, we leverage the global prototypes for rectification and impose additional constraints to promote the interclass separability and intraclass compactness. All prototypes are decoupled into a FG set and a background set to provide targeted segmentation cues. During the *meta-testing* phase, the FG set contains a FG prototype (fg) and a nonbase local prototype ($\mathbf{p}_{\text{local}}^{(0)}$), and the loss functions are not available.

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background set $\mathcal{P}_{\text{BG}}$ during the *meta-training* phase can be represented as

$$\mathcal{P}_{\text{FG}} = \left\{ \mathbf{p}_{\text{local}}^{(N_b+1)}, \mathbf{p}_{\text{merge}}^{(i)} \right\} \tag{5}$$

$$\mathcal{P}_{\text{BG}} = \left\{ \mathbf{p}_{\text{local}}^{(N_b+2)}, \left\{ \mathbf{p}_{\text{merge}}^{(i')} \right\}_{i' \in \mathcal{I}, i' \neq i} \right\} \tag{6}$$

where $|\mathcal{P}_{\text{FG}}| = 2$, $|\mathcal{P}_{\text{BG}}| = N_b + 1$, and $i \in \mathcal{I}$. During the *meta-testing* phase, the latter element $\mathbf{p}_{\text{merge}}^{(i)}$ in $\mathcal{P}_{\text{FG}}$ is replaced by $\mathbf{p}_{\text{local}}^{(0)}$, and the background set $\mathcal{P}_{\text{BG}}$ includes all the merged prototypes $\{\mathbf{p}_{\text{merge}}^{(i)}\}_{i \in \mathcal{I}}$. The category tackled by the current task changes from the base category to the novel category, and therefore the meaning of FG and background changes accordingly. In such a case, $|\mathcal{P}_{\text{FG}}| = 2$ and $|\mathcal{P}_{\text{BG}}| = N_b + 2$.

E. Decoupled Registration

With the decoupled prototype sets, we further utilize them to register query features and thus mine potential objects within the given image. The registration operation is conducted separately with respect to FG and background attributes to provide tailored guidance and prevent unnecessary confusion resulting from information redundancy. Taking the FG set $\mathcal{P}_{\text{FG}}$ as an example (see Fig. 4), each prototype is evaluated for pixel-level affinity with query features $\mathbf{F}_q$, and the generated activation features $\mathbf{F}_{\text{act}}$ can be defined as

$$\begin{aligned} \mathbf{F}_{\text{act}}^{(x,y,z)} &= \mathcal{C}\left(\mathbf{p}^z, \mathbf{F}_q^{(x,y)}\right) \\ &= \frac{\mathbf{p}^z \cdot \mathbf{F}_q^{(x,y)}}{\|\mathbf{p}^z\| \cdot \left\|\mathbf{F}_q^{(x,y)}\right\|} \end{aligned} \tag{7}$$

where $\mathbf{p}^z \in \mathcal{P}_{\text{FG}}$ indicates a specific prototype in the set. $\mathcal{C}(\cdot,\cdot)$ is the cosine metric used to measure the affinity of two vectors. (x, y) indexes all spatial locations and z indexes the prototype set. Note that the activation features regarding the background set $\mathcal{P}_{\text{BG}}$ can be computed in the same way.

Subsequently, an argmax operation along the z-dimension is implemented to determine the most appropriate prototype for providing guidance information. This calculation process can be mathematically defined as

$$z^\star = \mathcal{A}(\mathbf{F}_{\text{act}}, \text{dim} = z) \in \{0, \ldots, |\mathcal{P}_{\text{FG}}| - 1\}^{H \times W} \tag{8}$$

where $z^\star$ represents the indexes of the prototypes used for registration. $\mathcal{A}(\cdot, \text{dim} = z)$ denotes the argmax operation along the z-dimension.

Finally, we select the most correlated prototype for each location (x, y) according to the derived index $z^\star$, which is concatenated with the query features for registration and guidance. After dense comparison through the convolutional operation $\text{Conv}(\cdot)$, the query maps $\mathbf{M}_q$ are exploited to further filter irrelevant objects. The complete process described above can be expressed as follows:

$$\mathbf{F}_q^{\text{out}} = \mathbf{M}_q^{(0)} \cdot \text{Conv}\left(\mathbf{F}_\mathbf{q} \oplus \mathcal{T}\left(\mathcal{P}_{\text{FG}}, z^\star\right)\right) \tag{9}$$

where $\mathcal{T}(\cdot,\cdot)$ stands for indexing and stacking operations. $\text{Conv}(\cdot)$ represents a convolution with input and output channels of $2C$ and C, respectively. When performing background-related computations, the query maps for interference reduction become $\sum_{l=1}^{N_b} \mathbf{M}_q^{(l)}$ to highlight base-class regions.

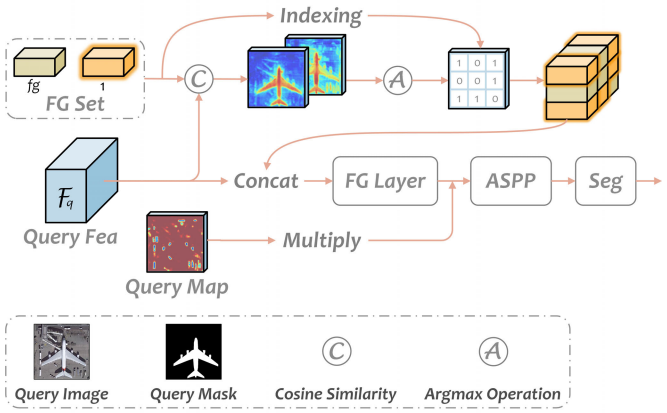


Fig. 4. Pipeline for DR during the *meta-training* phase. Each element in the prototype set is first evaluated the similarity score with the query feature, and the most correlated one is indexed for registration. These selected prototypes are then performed a point-wise concatenation with the query feature to propagate the object information. With the query map generated by the base layer, irrelevant objects can be effectively suppressed. Note that the FG set is served as an example here, and the procedure for the background one is similar.

F. Model Optimization

Our framework is trained in a mini-batch manner, where each metadata in the batch represents a specific episode. We employ the binary corss-entropy (BCE) as the primary loss function to evaluate the difference between the prediction mask and the ground-truth mask of the query image. Considering the complex characteristics of remote sensing images, we further introduce auxiliary losses (i.e., $\mathcal{L}_{\text{intra}}$ and $\mathcal{L}_{\text{inter}}$) to promote the intraclass compactness and interclass separability, which can be expressed as follows:

$$\mathcal{L}_{\text{intra}} = 1 - \frac{1}{N_b + 1} \sum_i \mathcal{C}\left(\mathbf{p}_{\text{local}}^{(i)}, \mathbf{p}_{\text{merge}}^{(i)}\right) \tag{10}$$

$$\mathcal{L}_{\text{inter}} = \frac{1}{N_b(N_b + 1)} \sum_i \sum_{i' \neq i} \mathcal{C}\left(\mathbf{p}_{\text{merge}}^{(i)}, \mathbf{p}_{\text{merge}}^{(i')}\right) \tag{11}$$

where $\mathcal{L}_{\text{intra}}$ measures the consistency of the prototype before and after rectification, while $\mathcal{L}_{\text{inter}}$ gauges the diversity among the rectified prototypes. $i \in \mathcal{I}$ indexes each prototype. Note that we explore the effect of loss weights on segmentation performance to achieve a trade-off between feature representations in a high-dimensional space.

Motivated by the superiority of scale-aware focal loss in [21], we meticulously design an analogous loss function $\mathcal{L}_{\text{scale}}$ to boost the model's capability to perceive potential objects and fit sample distributions, which can be defined as

$$\mathcal{L}_{\text{scale}} = -\left(1 - \hat{r}\right) \log\left(\hat{r}\right) - \delta^2 \log\left(1 - \delta^2\right) \tag{12}$$

$$r = \frac{1}{HW} \sum_{x,y} \mathbf{Y}_q^{(x,y)} \in [0, 1] \tag{13}$$

where $\hat{r}$ and r denote the proportion of FG regions in the prediction mask and the ground-truth mask of the query image, respectively. $\delta = \hat{r} - r$ indicates the difference between these two FG proportions. The former term in (12) allows the model to mine hard samples with under-segmentation, and the latter term constrains the probability distribution of output results

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TABLE I
NOVEL CLASSES FOR EACH FOLD OF iSAID-5^i DATASET

# Fold	Novel classes				
0	Ship (C1)	Storage tank (C2)	Baseball diamond (C3)	Tennis court (C4)	Basketball court (C5)
1	Ground track field (C6)	Bridge (C7)	Large vehicle (C8)	Small vehicle (C9)	Helicopter (C10)
2	Swimming pool (C11)	Roundabout (C12)	Soccer ball field (C13)	Plane (C14)	Harbor (C15)

as a whole. The weight coefficient of the scale-aware loss function $\mathcal{L}_{\text{scale}}$ is empirically set to 0.5 according to [21].

IV. EXPERIMENTS

A. Experimental Setup

1) Datasets: In recent years, an increasing number of studies on FSS have been witnessed, while its application in the field of remote sensing image processing is very few. For a fair comparison, we follow previous work [17] and construct a few-shot dataset based on the publicly available instance segmentation in aerial images dataset (iSAID) [12]. To be specific, iSAID is a large-scale and finely annotated segmentation dataset, which contains 6 55 451 object instances from 15 geospatial categories across 2806 high-resolution images. Under the few-shot setting, these geospatial categories are evenly divided into three folds: $\{5^i{:}i \in \{0, 1, 2\}\}$, where the novel categories corresponding to each fold are shown in Table I. The few-shot model is tested on a target fold i and trained on the remaining folds. Regarding the selection of novel categories within each fold, we follow the experimental setup of SDM [21], where the categories are determined according to the order of the label dictionary[1] in the original dataset.

2) Implementation Details: Prevalent FSS models [18], [19], [40] tend to use a backbone network pretrained on ImageNet [41] and then freeze its parameters during the *meta-training* phase to boost generalization ability. Considering the large domain shift between natural images and remote sensing images, we specifically train a segmentation network [39] with the standard learning paradigm on the base dataset to provide a more suitable backbone for the current task, where the learning rate is 1.25*e*-3, the batch size is 16, and the training epoch is 50. The entire segmentation network is optimized almost from scratch, except that the backbone part is initialized with parameters pretrained on ImageNet. The training data in this phase from the base dataset are consistent with those in the meta-training phase of the few-shot model, i.e., sampled from the base-class folds. It is worth noting that we reproduce the results of all related methods using such a strong backbone for a fair comparison (see Tables II–IV).

Our few-shot approach R^{2}Net is trained on the base dataset in an episodic fashion, where the learning rate, batch size, and training epoch are set to 0.05, 8, and 12, respectively. During the *meta-testing* phase, we randomly select 1000 episodes for evaluation and present the average results of 10 independent runs with different seeds. The raw images in the iSAID dataset are cropped to 512×512 for model training. The data augmentation techniques include horizontal flipping, Gaussian blurring, random rotation from $-10°$ to $10°$, and random scaling from 0.5 to 2.0. The number of channels C of the intermediate feature map is 256, and the base layer $\mathcal{B}(\cdot)$ is derived from the aforementioned standard segmentation network which is frozen during the *meta-training* stage. All experiments are implemented under the PyTorch framework and run on NVIDIA GeForce RTX 3090 GPUs.

All compared benchmark methods fall within the category of FSS and are implemented with parameter settings consistent with those recommended by the authors in their articles or open-source code. The hyper-parameters related to network training, such as learning rate, batch size, and number of epochs are set to the same values as our proposed R^{2}Net. We adopt this configuration to ensure the fairness of the experiment and the reliability of the results.

3) Evaluation Metrics: Following the common evaluation protocol of FSS [19], we adopt the mean intersection over union (mIoU) as our primary metric and the FG-background IoU (FB-IoU) as a secondary metric, where the former comprehensively integrates the performance of different categories and thus better reflects the segmentation capability. The mIoU metric can be calculated by: $\text{mIoU} = (1/N_n)\sum_{c=1}^{N_n}\text{IoU}_c$, where N_n represents the number of novel categories and IoU_c indicates the IoU score of category c. The FB-IoU metric can be computed by: $\text{FB-IoU} = (\text{IoU}_F + \text{IoU}_B)/2$ where IoU_F and IoU_B denote the FG and background IoU scores in the target fold i, respectively.

B. Results and Discussion

In this section, we compare the proposed R^{2}Net with mainstream FSS models. The results of all methods are presented in the same experimental setting to facilitate fair comparison.

1) Quantitative Analysis: The comparison results of our proposed method and other prevalent FSS frameworks in terms of mIoU metric are shown in Table II. We highlight the best results in bold and underline the second best ones for ease of comparison. On the whole, the proposed R^{2}Net achieves the best segmentation performance in the vast majority of cases. Such promising results demonstrate the unique advantages of our R^{2}Net in the few-shot scenario, which is not limited to a specific backbone or a specific number of shots. In particular, under the one-shot setting, our model outperforms the previous best performance by 2.52% with the VGG-16 backbone and 2.79% with the ResNet-50 backbone.

Among all the comparison methods, DMML [22], SDM [21], DML [45], and TBPN [46] are recently designed specifically for the FSS in remote sensing imagery. It can be noticed that the overall performance of our framework

[1]https://github.com/CAPTAIN-WHU/iSAID_Devkit/blob/master/preprocess/test_image_ids_json.py#L58

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TABLE II

PERFORMANCE COMPARISON ON iSAID-5^i IN MIOU UNDER ONE-SHOT AND FIVE-SHOT SETTINGS. "#PARAMS." DENOTES THE NUMBER OF LEARNABLE PARAMETERS. "FPS" INDICATES THE AVERAGED FRAME-PER-SECOND. RESULTS IN **BOLD** INDICATE THE BEST PERFORMANCE, WHILE THE UNDERLINED ONES REPRESENT THE SECOND BEST

Backbone	Method	1-shot				5-shot			
		Fold-0	Fold-1	Fold-2	Mean	Fold-0	Fold-1	Fold-2	Mean
VGG-16	$PANet_{2019}$ [42]	26.86	14.56	20.69	20.70	30.89	16.63	24.05	23.86
	$CANet_{2019}$ [18]	13.91	12.94	13.67	13.51	17.32	15.07	18.23	16.87
	SCL_{2021} [43]	25.75	18.57	22.24	22.19	35.77	**24.92**	**32.70**	31.13
	$PFENet_{2022}$ [19]	28.52	17.05	18.94	21.50	37.59	23.22	30.45	30.42
	$NERTNet_{2022}$ [40]	25.78	**20.01**	19.88	21.89	38.43	24.21	28.99	30.54
	DCP_{2022} [44]	28.17	16.52	22.49	22.39	39.65	22.68	29.93	30.75
	BAM_{2022} [35]	33.93	16.88	21.47	24.09	38.46	22.76	28.81	30.01
	$DMML_{2022}$ [22]	24.41	18.58	19.46	20.82	28.97	21.02	22.78	24.26
	SDM_{2022} [21]	24.52	16.31	21.01	20.61	26.73	19.97	26.10	24.27
	DML_{2022} [45]	30.99	14.60	19.05	21.55	34.03	16.38	26.32	25.58
	$TBPN_{2023}$ [46]	27.86	12.32	18.16	19.45	32.79	16.28	24.27	24.45
	R^2Net (*ours*)	**35.27**	19.93	**24.63**	**26.61**	**42.06**	23.52	30.06	**31.88**
ResNet-50	$PANet_{2019}$ [42]	27.56	17.23	24.60	23.13	36.54	16.05	26.22	26.27
	$CANet_{2019}$ [18]	25.51	13.50	24.45	21.15	29.32	21.85	26.91	26.03
	SCL_{2021} [43]	34.78	22.77	31.20	29.58	41.29	25.73	37.70	34.91
	$PFENet_{2022}$ [19]	35.84	23.35	27.20	28.80	42.42	25.34	33.00	33.59
	$NERTNet_{2022}$ [40]	34.93	**23.95**	28.56	29.15	44.83	26.73	37.19	36.25
	DCP_{2022} [44]	37.83	22.86	28.92	29.87	41.52	**28.18**	33.43	34.38
	BAM_{2022} [35]	39.43	21.69	28.64	29.92	43.29	27.92	38.62	36.61
	$DMML_{2022}$ [22]	28.45	21.02	23.46	24.31	30.61	23.85	24.08	26.18
	SDM_{2022} [21]	27.96	21.99	27.82	25.92	28.50	25.23	31.07	28.27
	DML_{2022} [45]	32.96	18.98	26.27	26.07	33.58	22.05	29.77	28.47
	$TBPN_{2023}$ [46]	29.33	16.84	25.47	23.88	30.98	20.42	28.07	26.49
	R^2Net (*ours*)	**41.22**	21.64	**35.28**	**32.71**	**46.45**	25.80	**39.84**	**37.36**

TABLE III

PERFORMANCE COMPARISON ON iSAID-5^i IN FB-IOU WITH RESNET-50 BACKBONE. "#PARAMS." AND "FPS" DENOTE THE NUMBER OF *Learnable* PARAMETERS AND THE AVERAGED FRAME-PER-SECOND UNDER ONE-SHOT SETTING, RESPECTIVELY

Method	1-shot	5-shot	#Params.	FPS
$PANet_{2019}$ [42]	56.56	57.37	23.6M	58.1
$CANet_{2019}$ [18]	56.64	59.46	22.3M	32.7
SCL_{2021} [43]	61.30	64.13	11.9M	39.2
$PFENet_{2022}$ [19]	60.09	63.25	10.8M	45.7
$NERTNet_{2022}$ [40]	59.97	64.45	20.8M	40.8
DCP_{2022} [44]	62.36	63.77	11.3M	37.9
BAM_{2022} [35]	62.04	65.00	**4.9M**	44.4
$DMML_{2022}$ [22]	57.78	58.26	23.6M	47.4
SDM_{2022} [21]	59.58	59.90	29.3M	52.9
DML_{2022} [45]	58.93	59.23	23.6M	**59.5**
$TBPN_{2023}$ [46]	57.34	58.63	23.6M	56.5
R^2Net (*ours*)	**63.86**	**66.18**	5.0M	41.5

is superior to theirs under various settings, establishing new state-of-the-arts in the current task. Considering that the scale-aware loss function in SDM [21] is applied to our R^2Net [see (12)], the performance difference between the two can be broadly attributed to other components, thus confirming the effectiveness of GR (Section III-D) and DR (Section III-E) from the side.

We also give the FB-IoU results for each method under different settings in Table III. As can be observed, our proposed R^2Net remains robust for this class-agnostic metric and significantly outperforms previous FSS methods. Taking the advanced performance of SDM [21] related to remote sensing as an example, our framework with ResNet-50 backbone achieves considerable gains of 4.28% and 6.28% in one-shot and five-shot settings, respectively. In addition, the penultimate column in Table III shows the number of learnable parameters for all methods during the meta-training phase. It can be found that our model enjoys a low optimization cost, with the "#Params." of ResNet-50 backbone being only 5.0 M. Table IV compares the class-wise performance of all methods under the one-shot setting. From left to right, the 15 geospatial classes (C1–C15) in groups of five indicate the segmentation accuracy on each fold of novel classes. It can be seen that our approach with VGG-16 backbone achieves the top-2 results in 8 out of 15 cases, while with ResNet-50 backbone only achieves 6 out of 15 cases. Focusing on specific categories, we observe that the proposed R^2Net performs better in ship (C1), storage tank (C2), tennis court (C4), and roundabout (C12) categories, but it faces challenges in accurately segmenting small vehicle (C9). This phenomenon is closely related to the interrelationships between categories and the scale of the instances to be determined. For example, large vehicles (C8) and small vehicles (C9) have similar semantics and appearances, making it quite challenging to differentiate their corresponding category prototypes based solely on scale differences. Bridges (C7) are located in highly diverse scenarios and often adjacent to roads, which imposes even greater demands on the discriminative ability of prototypes, particularly in terms of task relevance. Whereas categories such as storage tank (C2) and tennis court (C4) typically exhibit prominent features and are free from irrelevant distractions. The compact prototype representations of these categories facilitate accurate identification of objects.

2) Qualitative Analysis: Fig. 5 shows the visualization results of one-shot segmentation, where the baseline approach indicates our proposed R^2Net without GR, DR, and auxiliary loss function. It can be seen that the baseline scheme with our

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TABLE IV
CLASS-WISE PERFORMANCE COMPARISON IN MIOU UNDER ONE-SHOT SETTING USING VARIOUS BACKBONES

Method	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15	Mean
VGG-16																
$PANet_{2019}$ [42]	20.05	37.71	21.18	41.22	14.15	12.17	13.82	21.05	7.89	17.88	4.36	**31.68**	27.55	26.88	12.97	20.70
$CANet_{2019}$ [18]	24.13	6.73	13.83	16.32	8.54	14.12	3.24	21.04	3.35	22.96	9.57	14.91	17.83	16.11	9.92	13.51
SCL_{2021} [43]	28.50	32.93	19.68	29.60	18.05	22.48	7.92	31.46	8.99	22.02	14.17	16.53	19.72	**39.40**	21.37	22.19
$PFENet_{2022}$ [19]	34.32	31.81	24.20	35.43	16.86	13.98	6.01	31.68	6.76	**26.85**	8.15	17.75	20.56	33.34	14.87	21.50
$NERTNet_{2022}$ [40]	12.66	23.11	26.90	50.47	15.77	23.14	8.48	31.73	11.75	24.94	14.63	20.45	29.03	28.06	7.24	21.89
DCP_{2022} [44]	27.69	38.45	25.92	33.20	15.57	17.62	12.36	26.79	8.05	17.80	**22.45**	18.29	18.03	37.57	16.10	22.39
BAM_{2022} [35]	27.66	43.90	**31.48**	43.96	**22.66**	13.57	8.91	**31.76**	9.26	20.91	17.05	26.27	**30.68**	25.27	8.07	24.09
$DMML_{2022}$ [22]	34.75	37.36	15.15	22.85	11.94	21.41	13.85	23.92	10.24	23.50	8.17	16.32	21.08	29.63	**22.09**	20.82
SDM_{2022} [21]	33.76	23.88	17.80	27.76	19.38	18.36	9.63	25.24	8.63	19.69	10.56	15.36	24.76	32.30	22.06	20.61
DML_{2022} [45]	27.30	42.63	19.25	**50.63**	15.13	14.16	**15.94**	22.40	7.74	12.74	3.79	23.73	23.47	27.40	16.88	21.55
$TBPN_{2023}$ [46]	22.03	39.75	20.80	42.80	13.94	10.41	6.87	16.54	4.38	23.41	5.68	23.66	22.13	24.63	14.72	19.45
R^2Net (*ours*)	**37.82**	**45.16**	26.27	45.30	21.81	**24.11**	14.38	30.92	**12.21**	18.03	18.66	25.02	29.64	31.95	17.87	**26.61**
ResNet-50																
$PANet_{2019}$ [42]	21.81	36.31	23.01	42.06	14.59	12.11	17.44	22.70	12.27	21.60	30.29	24.62	26.79	25.54	15.79	23.13
$CANet_{2019}$ [18]	39.57	18.54	18.46	33.63	17.34	9.78	5.49	22.15	5.17	24.89	9.96	36.50	19.12	38.82	17.85	21.15
SCL_{2021} [43]	37.61	33.63	26.68	54.75	21.22	22.60	24.40	30.22	6.71	29.93	**33.00**	44.68	18.25	**44.63**	15.46	29.58
$PFENet_{2022}$ [19]	39.02	45.63	20.86	49.96	23.72	21.00	24.76	31.59	6.98	32.42	13.34	**47.64**	30.65	32.82	11.54	28.80
$NERTNet_{2022}$ [40]	33.59	42.83	22.30	49.35	21.91	21.62	**28.82**	25.64	9.35	34.30	23.91	38.67	25.63	40.84	13.74	28.83
DCP_{2022} [44]	37.42	42.44	35.16	56.55	17.58	21.66	19.57	32.97	10.60	29.50	24.02	35.34	28.44	39.80	17.02	29.87
BAM_{2022} [35]	36.34	39.76	**38.23**	**58.13**	24.71	18.25	12.68	**35.91**	11.42	30.21	28.98	40.74	29.43	33.25	10.79	29.92
$DMML_{2022}$ [22]	40.14	40.18	21.31	27.02	13.60	15.56	15.19	26.05	13.84	**34.44**	11.26	17.57	23.27	39.11	**26.12**	24.31
SDM_{2022} [21]	41.77	35.50	21.41	20.81	20.29	15.60	25.60	28.66	13.29	26.79	13.61	32.35	24.59	42.79	25.75	25.92
DML_{2022} [45]	35.13	42.10	30.49	41.79	15.31	13.25	16.87	24.70	**14.62**	25.45	10.24	35.49	25.35	41.69	18.57	26.07
$TBPN_{2023}$ [46]	25.36	41.28	30.67	32.88	16.48	13.48	9.74	27.88	12.52	20.56	11.12	34.31	23.57	40.36	17.98	23.88
R^2Net (*ours*)	**46.87**	**49.06**	30.70	52.86	**26.62**	**24.31**	17.25	31.25	13.67	21.73	24.88	46.07	**42.29**	42.07	21.08	**32.71**

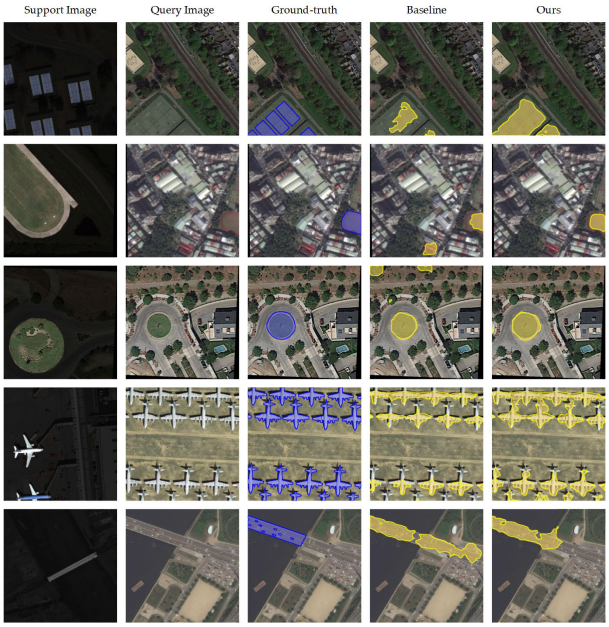


Fig. 5. Visualization of FSS results. From left to right, each image in a specific row represents the support image, query image, ground-truth, baseline result, and our result, respectively. It can be noticed that the proposed R^2Net can improve the segmentation performance comprehensively (e.g., precision and recall) and avoid the interference from irrelevant objects to a certain extent.

proposed strategies can noticeably improve the accuracy of geospatial object segmentation and mitigate interclass confusion from a global perspective. Fig. 6 depicts the support maps $\mathbf{M_s}$ generated by the base layer $\mathcal{B}(\cdot)$ on different folds [see (1)], and the accurate segmentation of geospatial objects implies the reliability of leveraging them to suppress interference.

C. Ablation Studies

1) Ablation Studies of the proposed R^2Net at the Module Level: In order to verify the effectiveness of the proposed GR and DR, we conduct ablation studies at the module level.

TABLE V
ABLATION STUDIES OF THE PROPOSED R^2NET AT THE MODULE LEVEL. "GR" AND "DR" DENOTE THE GR AND DR MODULES, RESPECTIVELY

GR	DR	Fold-0	Fold-1	Fold-2	Mean
		31.62	17.11	20.34	23.02
✓		34.92	18.14	22.51	25.19
	✓	34.66	18.42	23.67	25.58
✓	✓	**35.27**	**19.93**	**24.63**	**26.61**

It can be observed from Table V that both GR and DR modules have made positive contributions to the segmentation performance. The average mIoU gains achieved over all folds compared to the baseline scheme (first row) are 2.17% and 2.56% for the GR and DR modules, respectively. We attribute these impressive results to the strong generalization ability of the global prototype and the targeted nature of the decoupled cues. Simultaneously, the two modules complement each other, further enhancing the inference accuracy of the few-shot model.

2) Ablation Studies of the Fusion Method for Global and Local Prototypes: As discussed in Section III-D, global and local prototypes can be merged by either fixed or learnable coefficients. Table VI shows the comparison of the segmentation results for the two schemes. It can be found that the scheme with learnable coefficients performs better by virtue of its adaptability and flexibility. The last two rows in Table VI illustrate that utilizing the query map to suppress interference does have a positive effect. Fig. 7 compares the changes in interclass and intraclass relationships during the training process of different variants, where the learnable scheme clearly improves the discriminability of feature space.

3) Ablation Studies of the Segmentation Head: The decoupled FG and background information flows are fed separately into the segmentation head and then reintegrated to produce the prediction mask. To prevent feature confusion, we attempt

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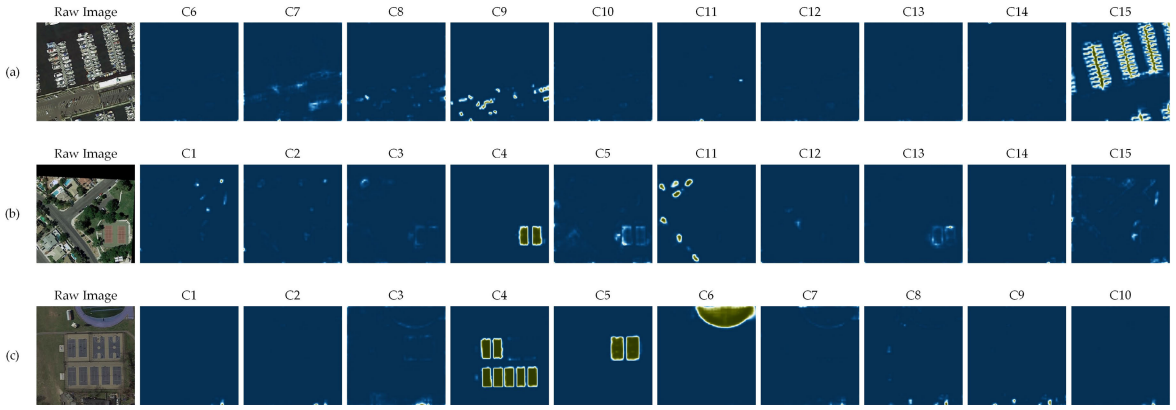


Fig. 6. Visualization results of the support maps generated by the base layer $\mathcal{B}(\cdot)$. Each row from (a)–(c) corresponds to the results of Fold-0 to Fold-2, respectively. There are ten feature maps depicted since the iSAID-5^i dataset is divided into ten base classes and five novel classes.

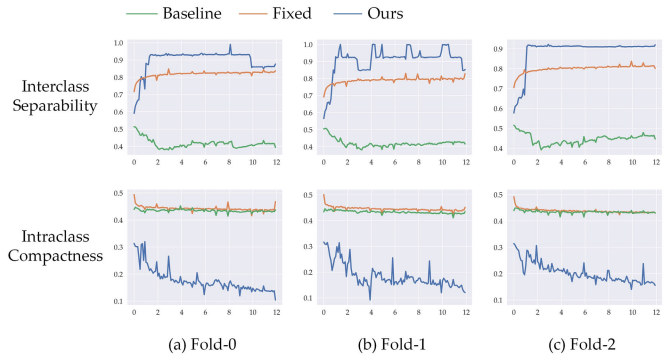


Fig. 7. Convergence curves for interclass separability and intraclass compactness across various folds. "Fixed" indicates that the global and local prototypes are merged using a fixed weight of 0.5. (a) Fold-0. (b) Fold-1. (c) Fold-2.

to leverage two independent segmentation heads to process the decoupled information. The experimental results in Table VII indicate that: 1) the shared segmentation head enjoys higher accuracy and efficiency and 2) using a one-layer network to encode FG - or background-specific cues is sufficient, as the "FG Layer" in Fig. 4.

4) Ablation Studies of the Loss Function: Table VIII presents the mIoU results of the baseline schemes with different auxiliary loss functions under the one-shot setting. It can be found that introducing $\mathcal{L}_{\text{intra}}$ obtains greater performance gains compared to introducing $\mathcal{L}_{\text{inter}}$, e.g., 25.58% versus 22.88% mIoU scores, indicating the importance of intraclass compactness for the proposed framework. The inferior performance exhibited by the fourth variant (w/ $\mathcal{L}_{\text{inter}}$ and w/o $\mathcal{L}_{\text{intra}}$) can be attributed to the fact that focusing solely on interclass relationships without considering intraclass compactness may lead to overlapping distributions of different categories within the feature space. As a consequence, semantic confusion occurs among the extracted prototype representations, ultimately compromising the segmentation capability. Taken as a whole, these three losses exhibit a complementary relationship, and the variant employing them in combination achieves the highest accuracy (see the last row). Moreover, Fig. 8 shows the ablation results on the weights of loss functions, where the optimal performance is obtained at $(0.25, 0.25)$.

TABLE VI

ABLATION STUDIES OF THE FUSION METHOD FOR GLOBAL AND LOCAL PROTOTYPES. THE QUERY MAP $\mathbf{M}_q^{(0)}$ FURTHER SUPPRESSES IRRELEVANT REGIONS FOR THE FUSED PROTOTYPE. THE LAST ROW REPRESENTS THE BASELINE SCHEME USING ONLY LOCAL PROTOTYPES

Learnable	Fixed	$\mathbf{M}_q^{(0)}$	Fold-0	Fold-1	Fold-2	Mean
	✓	✓	**35.61**	16.73	21.22	24.52
✓		✓	35.27	**19.93**	**24.63**	**26.61**
✓			33.67	19.44	24.13	25.75
–	–		32.37	17.00	20.89	23.42

TABLE VII

ABLATION STUDIES OF THE SEGMENTATION HEAD. "FPS" DENOTES THE AVERAGED FRAME-PER-SECOND

Shared Head	#Params.	FPS	mIoU	FB-IoU
✓	4.7M	84.7	26.61	60.28
✗	8.7M	65.4	25.34	59.40

TABLE VIII

ABLATION STUDIES OF THE LOSS FUNCTION

$\mathcal{L}_{\text{inter}}$	$\mathcal{L}_{\text{intra}}$	$\mathcal{L}_{\text{scale}}$	Fold-0	Fold-1	Fold-2	Mean
		✓	30.79	16.48	22.48	23.25
✓	✓		31.50	19.89	19.07	23.49
	✓	✓	35.22	18.20	23.31	25.58
✓		✓	33.74	18.75	16.16	22.88
✓	✓	✓	**35.27**	**19.93**	**24.63**	**26.61**

5) Ablation Studies of the Five-Shot Fusion Approach: When more support samples as provided, i.e., the K-shot scenario, it is common practice to average the prototype representations from multiple samples for segmentation guidance. In this work, we provide another simple yet powerful alternative that performs a weighted fusion according to the affinity score of each prototype to query features. Table IX compares the results of fusion approaches with different backbones.

6) Ablation Studies of the Dataset Division Scheme: Similar to the two different division methods of PANet [42] and FWB [47] for the COCO-20^i benchmark in the standard FSS, we also evaluate the effectiveness of our proposed R^2Net under another division scheme. Specifically, in the division

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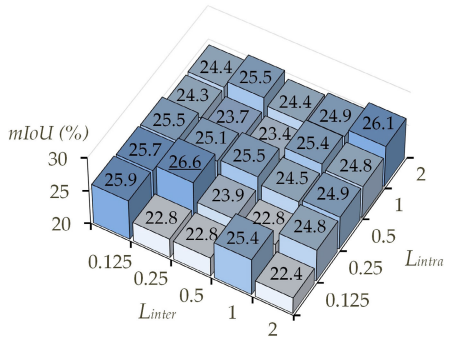


Fig. 8. Ablation studies on the weights of loss functions $\mathcal{L}_{\text{inter}}$ and $\mathcal{L}_{\text{intra}}$.

TABLE IX
ABLATION STUDIES OF THE FIVE-SHOT FUSION APPROACH

Method	VGG-16				ResNet-50			
	Fold-0	Fold-1	Fold-2	Mean	Fold-0	Fold-1	Fold-2	Mean
Mean	41.64	21.06	28.72	30.47	43.96	23.15	38.40	35.17
Softmax	42.06	23.52	30.06	31.88	46.45	25.80	39.84	37.36

TABLE X
ABLATION STUDIES OF THE DATASET DIVISION SCHEME

Method	Split	1-shot				5-shot			
		Fold-0	Fold-1	Fold-2	Mean	Fold-0	Fold-1	Fold-2	Mean
$DMML_{2022}$ [22]	A	24.41	18.58	19.46	20.82	28.97	21.02	22.78	24.26
SDM_{2022} [21]	A	24.52	16.31	21.01	20.61	26.73	19.97	26.10	24.27
R^2Net (*ours*)	A	**35.27**	**19.93**	**24.63**	**26.61**	**42.06**	**23.52**	**30.06**	**31.88**
$DMML_{2022}$ [22]	B	25.59	23.96	16.37	21.97	27.99	26.33	19.70	24.67
SDM_{2022} [21]	B	20.07	24.73	18.05	20.95	22.67	30.94	21.03	24.88
R^2Net (*ours*)	B	**27.09**	**35.61**	**18.87**	**27.19**	**32.78**	**41.27**	**21.56**	**31.87**

scheme described in Table I (denoted as Split-A), the category indices corresponding to each fold are: $\{i + 3j + 1\}$ where $i \in \{0, 1, 2\}$, $j \in \{0, 1, \ldots, 4\}$; while in the newly introduced division scheme (denoted as Split-B), the class indices are: $\{5i + j + 1\}$. It can be observed from Table X that our R^2Net demonstrates significant advantages over state-of-the-art remote sensing FSS models with different division ways, implying the reliability of the achieved performance gains.

V. CONCLUSION

Most of the current FSS studies focus on natural image processing, and few of them are specifically designed considering the complex characteristics of remote sensing images. To this end, we propose a novel and powerful framework with GR and DR, which improves the accuracy of prototype activation and avoids unnecessary confusion from information redundancy. Furthermore, the introduction of ancillary losses promotes interclass separability and intraclass compactness. The proposed approach, termed R^2Net, is comprehensively evaluated on the standard benchmark iSAID-5^i and establishes new state-of-the-art in multiple few-shot settings.

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